HW1

1 Question 1

Open Science Collaboration (2015) attempted to replicate 100 psychology experiments from three journals:

- Journal of Personality and Social Psychology (JPSP)
- Journal of Experimental Psychology: Learning, Memory, and Cognition (JEP: LMC).
- Psychological Science (PSCI)

Journal	Replicated	Not Replicated
JPSP	7	24
JEP:LMC	13	14
PSCI	15	24

Their work astonishingly showed that, empirically, only just over half of their attempts were successful.

1.1 Part a

Overall, is there support for the claim that more than half of psychology research is able to be replicated?

```
library(stats)
library(pwr)
library(binom)

replicated = 7 + 13 + 25
not_replicated = 24 + 14 + 24
total = replicated + not_replicated

binom.test(replicated, total, p = 0.5, alternative = "greater")
```

```
    Exact binomial test

data:  replicated and total
number of successes = 45, number of trials = 107, p-value = 0.9593
alternative hypothesis: true probability of success is greater than 0.5
95 percent confidence interval:
 0.339836 1.000000
sample estimates:
probability of success
             0.4205607
```

```
ES.h(0.5, replicated/total)
```

```
[1] 0.1595546
```

```
binom.exact(replicated, total)
```

```
  method  x   n      mean    lower     upper
```

```
1  exact 45 107 0.4205607 0.325769 0.5198639
```

Based on an exact binomial test, there is not support for the claim that over half of psychology research is able to be replicated. We fail to reject our null hypothesis that $p = 0.5$ (p-value $= 0.9593$). A 95% confidence interval for our proportion is [0.33, 0.52]. Our effect size, cohen's h, is 0.160, indicating a small difference between our empirical proportion of 0.421 and our null proportion of 0.5.

1.2 Part b

Assess whether the replication success rate varies across journals.

```
library(effectsize)

replication_dat <- tibble(Replicated = c(7, 13, 15),
                          Not_Replicated = c(24, 14, 24))
rownames(replication_dat) <- c("JPSP", "JEP:LMC", "PSCI")

chisq.test(replication_dat)
```

```
    Pearson's Chi-squared test

data:  replication_dat
X-squared = 4.2504, df = 2, p-value = 0.1194
```

```
cramers_v(replication_dat)
```

```
Cramer's V (adj.) |       95% CI
---------------------------------
0.15              | [0.00, 1.00]

- One-sided CIs: upper bound fixed at [1.00].
```

```
interpret_cramers_v(0.15)
```

```
[1] "small"
(Rules: funder2019)
```

Based on the results from a Chi-Squared test of independence, there is no evidence for a statistically discernible difference in replication rates by journal ($\chi^2 = 4.250$, $p = 0.119$). The effect size is small (Cramer's $v = 0.15$).

2 Question 2

Below, I simulate a rather quite annoying set of regression data.

```
set.seed(0510)
n <- 250
dat <- tibble(x1 = rnorm(n, mean = 50, sd = 10),
              x2 = factor(sample(c("A", "B"), n, replace = TRUE)),
              x3 = factor(sample(c("Control", "Treat"), n, replace = TRUE)),
              x4 = factor(sample(c("Low", "High"), n, replace = TRUE)),
              x5 = rnorm(n, mean = 0, sd = 1),
              x6 = factor(sample(c("Group1", "Group2"), n, replace = TRUE)),
              x7 = rnorm(n, mean = 10, sd = 2),
              x8 = rnorm(n, mean = 10, sd = 2)) |>
       mutate(y = 5 +                          # Intercept
                  # Quantitative x1
                  2 * x1 +
                  # Categorical x2
                  (x2 == "B") * 10 +
                  # Categorical x Categorical (x3*x4)
                  (x3 == "Treat") * 5 + (x4 == "High") * 2 +
                  (x3 == "Treat" & x4 == "High") * 15 +
                  # Quantitative x Categorical  (x5*x6)
                  3 * x5 + (x6 == "Group2") * 4 +
                  (x5 * (x6 == "Group2")) * 8 +
                  # Quantitative x Quantitative (x7*x8)
                  1.5 * x7 + 0.5 * x8 +
```

```
                0.8 * (x7 * x8) +
                # Error
                rnorm(n, mean = 0, sd = 15))
```

2.1 Part a

Fit the regression model.

$$
\begin{aligned}
Y_i = &\beta_0 + \beta_1 x_{1i} + \beta_2 I(x_{2i} = \text{B}) + \beta_3 I(x_{3i} = \text{Treat}) + \beta_4 (x_{4i} = \text{High}) + \\
&\beta_5 x_{5i} + \beta_6 I(x_{6i} = \text{Group 2}) + \beta_7 x_{7i} + \beta_8 x_{8i} + \\
&\beta_9 (I(x_{3i} = \text{Treat}) \times I(x_{4i} = \text{High})) + && \text{(Categorical} \times \text{Categorical)} \\
&\beta_{10} (x_{5i} \times I(x_{6i} = \text{Group 2})) + && \text{(Continuous} \times \text{Categorical)} \\
&\beta_{11} (x_{7i} \times x_{8i}) + \epsilon_i && \text{(Continuous} \times \text{Continuous)}
\end{aligned}
$$

```
formula = y ~ x1 + x2 + x3 * x4 + x5 * x6 + x7 * x8

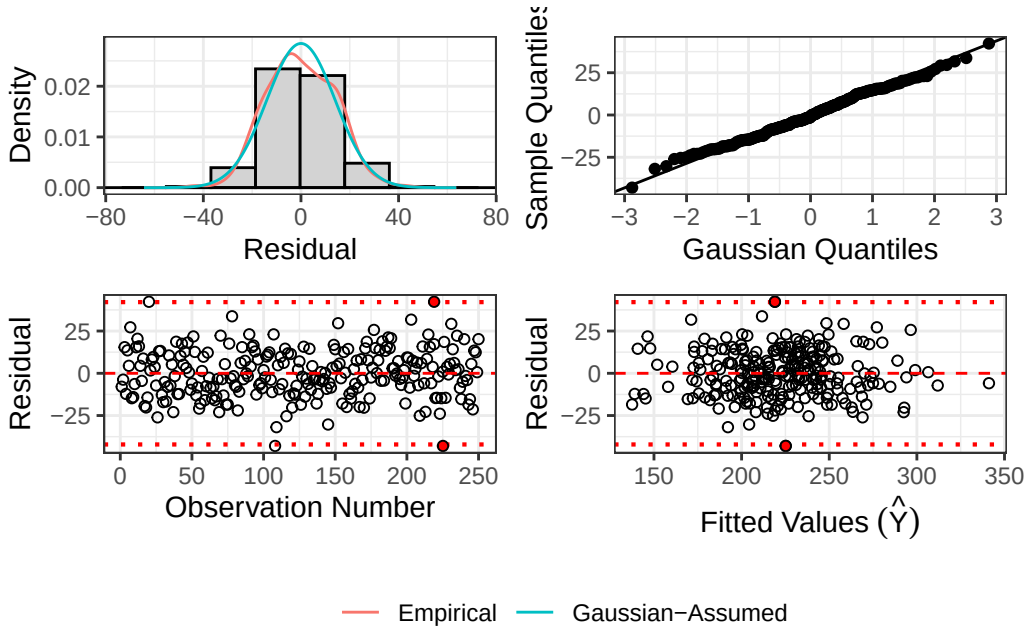
mod <- lm(formula, dat)
```

2.2 Part b

Conduct a residual and outlier analysis.

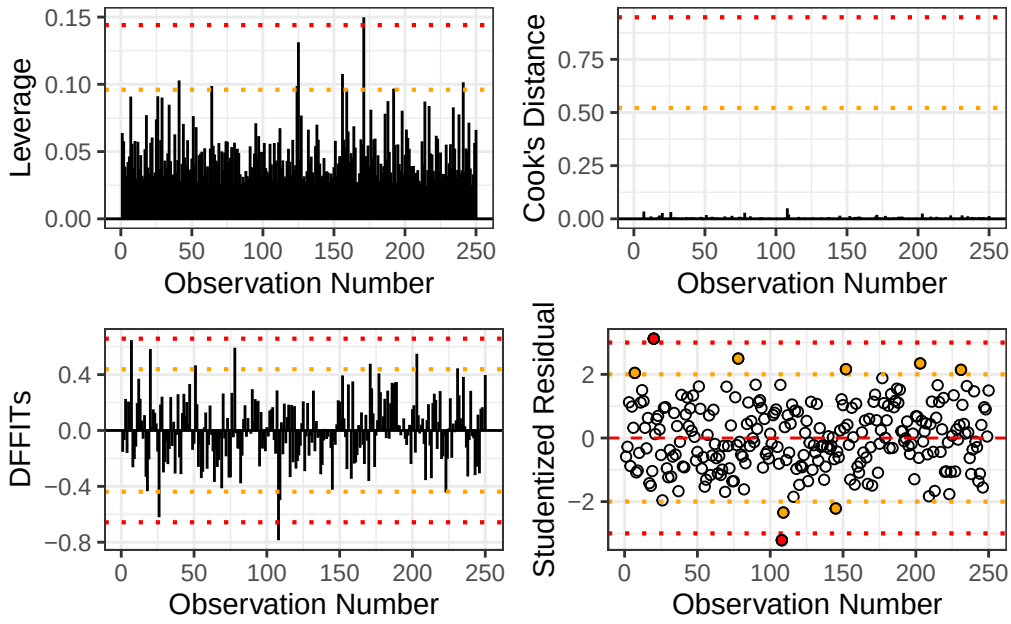
```
library(ggplot2)
library(patchwork)
source(file = "plotResiduals.R")

plotResiduals(mod)
```



A histogram of our linear model residuals shows that the residuals appear very normal and are centered around 0. The qq-plot of the residuals shows a very close fit between the Gaussian quantiles and sample quantiles. There are two residuals that are on the border of reasonability but that alone shouldn't be much concern.

```
source(file = "plotInfluence.R")
plotInfluence(mod)
```



There is one observation with a high leverage, one point with a large negative DFFITs, and two observations with large positive or negative studentized residuals. Given that none of the observations have high Cook's D values, only one observation is flagged in two metrics, and the total sample size is large, these metrics on the influence of outliers appear fairly standard to a true linear model.

2.3 Part c

Interpret the intercept, the coefficients on x_1 and x_2.

```
1 summary(mod)
```

```
Call:
lm(formula = formula, data = dat)

Residuals:
    Min      1Q  Median      3Q     Max
-42.979  -9.395  -1.054  10.150  42.305

Coefficients:
               Estimate Std. Error t value Pr(>|t|)
(Intercept)    12.41835   26.71681   0.465 0.642489
x1              2.13546    0.08944  23.876  < 2e-16 ***
x2B            10.02851    1.79959   5.573 6.77e-08 ***
x3Treat        21.70357    2.54065   8.543 1.59e-15 ***
x4Low          -3.52182    2.53119  -1.391 0.165413
x5              2.48350    1.29418   1.919 0.056183 .
x6Group2        4.98495    1.80989   2.754 0.006336 **
x7              0.37174    2.61136   0.142 0.886920
x8             -0.92790    2.60484  -0.356 0.721990
x3Treat:x4Low -15.19194    3.66722  -4.143 4.77e-05 ***
x5:x6Group2     6.94743    1.85460   3.746 0.000225 ***
x7:x8           0.92871    0.25937   3.581 0.000415 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 14.03 on 238 degrees of freedom
Multiple R-squared:  0.8562,    Adjusted R-squared:  0.8495
F-statistic: 128.8 on 11 and 238 DF,  p-value: < 2.2e-16
```

The intercept tells us that for a point with all numeric variables (x_1, x_5, x_7, x_8) equalling 0, x_2 = "A", x_3 = "Control", x_4 = "High", and x_6 = "Group 1", the model would predict a y value of 12.418.

The x_1 coefficient tells us that for every unit increase in x_1, we'd expect a y value increase of 2.135.

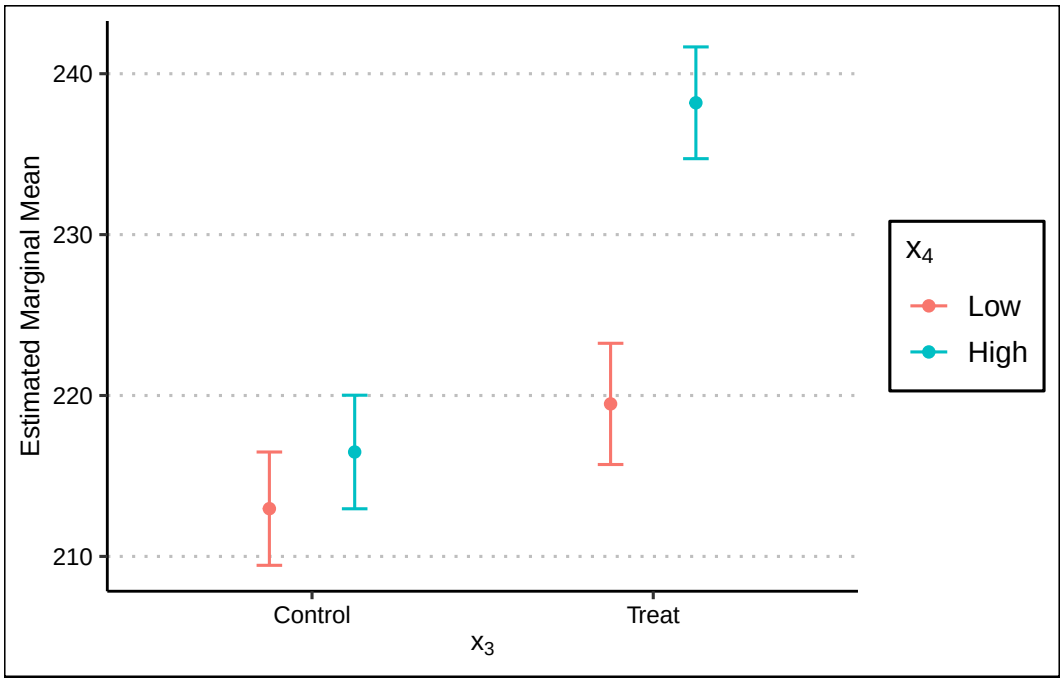
The x_2 coefficient tells us that being in the "B" group is associated with a 10.029 increase in y compared to the "A" group.

2.4 Part d

Untangle and interpret the $x_3 * x_4$ interaction.

```
library(ggthemes)
library(emmeans)
mod.emm <- summary(emmeans(mod, ~ x4 | x3))
mod.emm$x4 = factor(mod.emm$x4, levels = c("Low", "High"))

ggplot(data = mod.emm, aes(x = x3, y = emmean, color = x4)) +
  geom_point(position = position_dodge(width = 0.5))+
  geom_errorbar(aes(ymin = lower.CL, ymax = upper.CL), width = 0.15, position = position_dodge(width = 0.5)
  xlab(bquote(x[3])) +
  labs(color = bquote(x[4])) +
  ylab("Estimated Marginal Mean") +
  theme_clean()
```




```
mod.pairs <- emmeans(mod, ~ x4 | x3)
pairs(mod.pairs)
```

```
x3 = Control:
 contrast   estimate   SE  df t.ratio p.value
 High - Low     3.52 2.53 238   1.391  0.1654

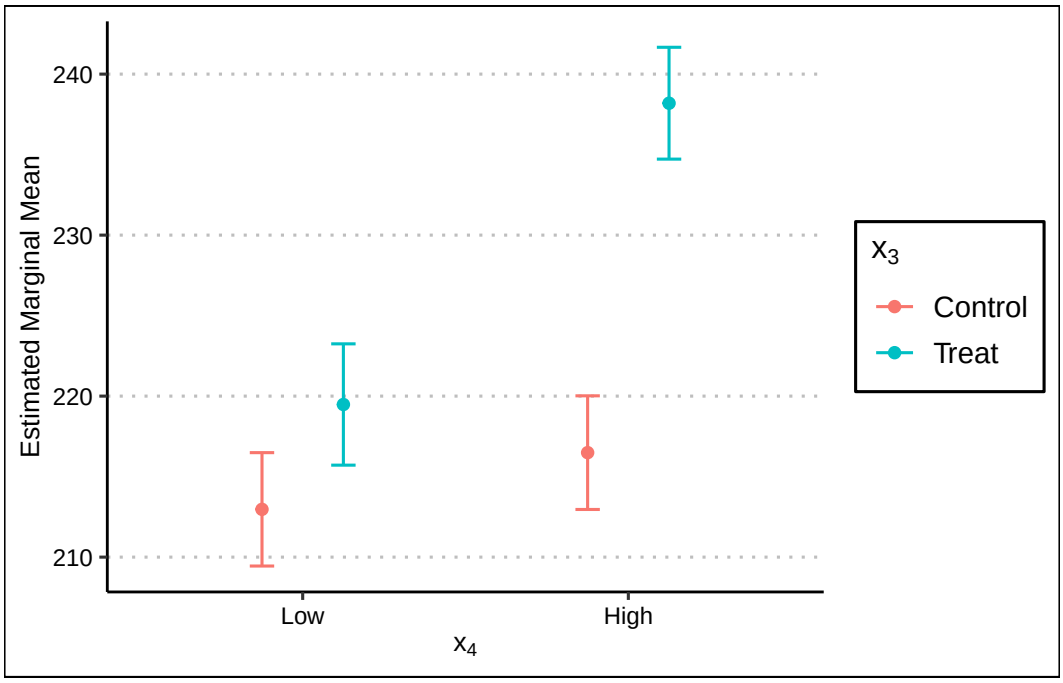
x3 = Treat:
 contrast   estimate   SE  df t.ratio p.value
```

```
 High - Low    18.71 2.60 238   7.195  <.0001

Results are averaged over the levels of: x2, x6
```

If we view the effect of x_4 across the levels of x_3, we see that in the control group, having x_4 = "High" was associated with a 3.52 greater value in y as compared to x_4 "Low", a difference not statistically discernible ($p = 0.165$). In the treat group, having x_4 = "High" was associated with 18.71 greater value in y as compared to x_4 = "Low", a statistically discernible difference ($p < 0.0001$).

```
ggplot(data = mod.emm, aes(x = x4, y = emmean, color = x3)) +
  geom_point(position = position_dodge(width = 0.5))+
  geom_errorbar(aes(ymin = lower.CL, ymax = upper.CL), width = 0.15, position = position_dodge(width = 0.5)
  xlab(bquote(x[4])) +
  labs(color = bquote(x[3])) +
  ylab("Estimated Marginal Mean") +
  theme_clean()
```




```
mod.pairs <- emmeans(mod, ~ x3 | x4)
pairs(mod.pairs)
```

```
x4 = High:
 contrast        estimate   SE  df t.ratio p.value
 Control - Treat   -21.70 2.54 238  -8.543  <.0001

x4 = Low:
 contrast        estimate   SE  df t.ratio p.value
 Control - Treat    -6.51 2.66 238  -2.445  0.0152

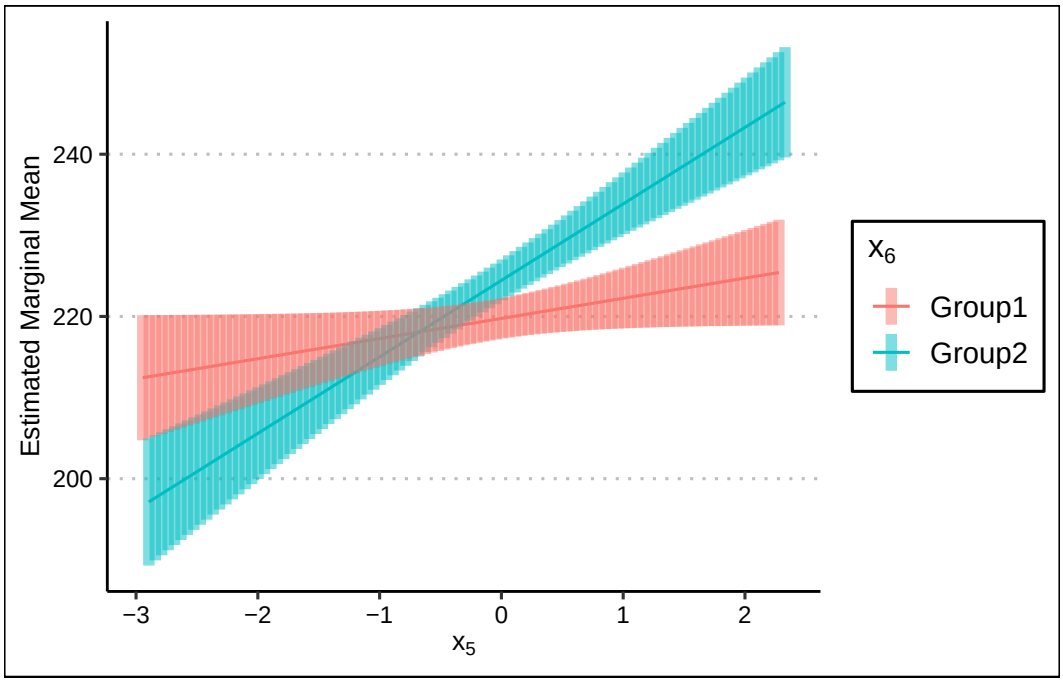
Results are averaged over the levels of: x2, x6
```

If we view the effect of x_3 across the levels of x_4, we see that in the low group, having x_3 = "Treat" was associated with 6.51 greater value in y as compared to x_3 = "Control", a statistically discernible difference ($p = 0.015$). In the high group, having x_3= "Treat" was associated with a 21.70 greater value in y as compared to x_3 = "Control", a statistically discernible difference ($p < 0.0001$).

2.5 Part e

Untangle and interpret the $x_5 * x_6$ interaction.

```
emmip(mod, x6 ~ x5,
      at = list(x5 = seq(min(dat$x5), max(dat$x5), length.out = 100)),
      CIs = TRUE) +
  ylab("Estimated Marginal Mean") +
  xlab(bquote(x[5])) +
  labs(color = bquote(x[6])) +
  theme_clean()
```



From the interaction plot we see that the slope of x_5 appears to be greater for the Group 2 in x_6 than Group 1.

```
emtrends(mod, "x6", var = "x5")
```

```
 x6     x5.trend   SE  df lower.CL upper.CL
 Group1     2.48 1.29 238   -0.066     5.03
 Group2     9.43 1.34 238    6.798    12.06

Results are averaged over the levels of: x2, x3, x4
Confidence level used: 0.95
```

The trend of x_5 in Group 1 is 2.48 (95% CI: [-0.066, 5.03]), while the trend of x_5 in Group 2 is 9.43 (95% CI: [6.798, 12.06]).

```
pairs(emtrends(mod, "x6", var = "x5"))
```

```
 contrast        estimate   SE  df t.ratio p.value
 Group1 - Group2    -6.95 1.85 238  -3.746  0.0002

Results are averaged over the levels of: x2, x3, x4
```

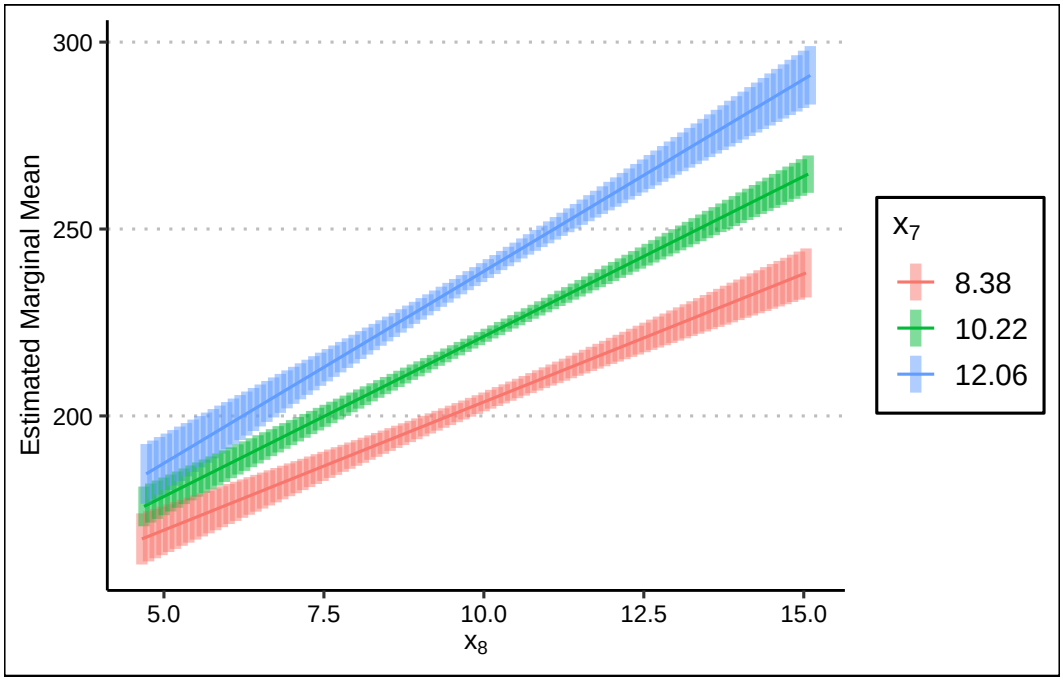
The trend of x_5 for Group 2 is 6.95 greater than the trend for Group 1, a difference that is statistically discernible ($p = 0.0002$).

2.6 Part f

Untangle and interpret the $x_7 * x_8$ interaction.

```r
x7.vals <- c(
  low  = mean(dat$x7) - sd(dat$x7),
  mean = mean(dat$x7),
  high = mean(dat$x7) + sd(dat$x7)
)

emmip(mod, x7 ~ x8,
      at = list(
        x8 = seq(min(dat$x8), max(dat$x8), length.out = 100),
        x7 = round(x7.vals, 2)
      ),
      CIs = TRUE) +
  xlab(bquote(x[8])) +
  ylab("Estimated Marginal Mean") +
  labs(color = bquote(x[7]), fill = bquote(x[7])) +
  theme_clean()
```




```r
mod.emt <- emtrends(mod, ~ x7, var = "x8", at = list(x7 = round(x7.vals,2)))

pairs(mod.emt)
```

```
 contrast          estimate    SE  df t.ratio p.value
 x78.38 - x710.22     -1.71 0.477 238  -3.581  0.0012
 x78.38 - x712.06     -3.42 0.954 238  -3.581  0.0012
 x710.22 - x712.06    -1.71 0.477 238  -3.581  0.0012

Results are averaged over the levels of: x2, x3, x4, x6
P value adjustment: tukey method for comparing a family of 3 estimates
```

Trends for x_8 are discernibly different at low, average, and high (+/-1 SD) levels of x_7, increasing by 1.71 at each level (all p-values = 0.0012).

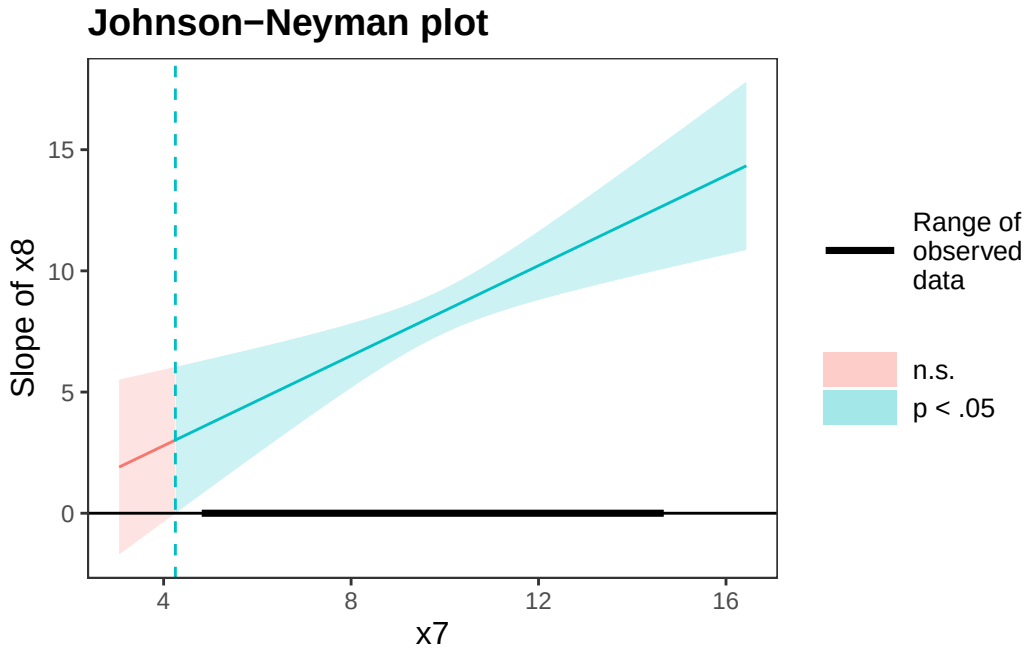
```r
library(interactions)

johnson_neyman(mod, pred = x8, modx = x7)
```

```
JOHNSON-NEYMAN INTERVAL

When x7 is OUTSIDE the interval [-9.96, 4.25], the slope of x8 is p < .05.

Note: The range of observed values of x7 is [4.89, 14.60]
```



A Johnson-Neyman plot shows us that the slope of x_8 increases as x_7 increases and is statistically discernible ($p < 0.05$) for the range of observed data. For theoretical lower levels of x_7 (<4.25), the slope of x_8 would not be statistically discernible ($p \geq 0.05$).

2.7 Part g

Standardize the continuous data and refit the regression model. What do you notice about the coefficient estimates in relation to your answers to parts d-f?

```r
dat_scaled <- dat |>
  mutate(x1 = scale(x1),
         x5 = scale(x5),
         x7 = scale(x7),
         x8 = scale(x8),
         y = scale(y))

mod_scaled <- lm(formula, dat_scaled)
summary(mod_scaled)
```

```
Call:
lm(formula = formula, data = dat_scaled)

Residuals:
     Min       1Q   Median       3Q      Max
-1.18783 -0.25965 -0.02912  0.28052  1.16921

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)   -0.35574    0.06272  -5.672 4.08e-08 ***
```

```
x1             0.60221    0.02522  23.876  < 2e-16 ***
x2B            0.27716    0.04974   5.573 6.77e-08 ***
x3Treat        0.59983    0.07022   8.543 1.59e-15 ***
x4Low         -0.09733    0.06996  -1.391 0.165413
x5             0.06803    0.03545   1.919 0.056183 .
x6Group2       0.12472    0.04985   2.502 0.013019 *
x7             0.49350    0.02548  19.369  < 2e-16 ***
x8             0.45617    0.02527  18.052  < 2e-16 ***
x3Treat:x4Low -0.41987    0.10135  -4.143 4.77e-05 ***
x5:x6Group2    0.19030    0.05080   3.746 0.000225 ***
x7:x8          0.09094    0.02540   3.581 0.000415 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3879 on 238 degrees of freedom
Multiple R-squared:  0.8562,    Adjusted R-squared:  0.8495
F-statistic: 128.8 on 11 and 238 DF,  p-value: < 2.2e-16
```

Most of the p-values for the coefficients have stayed the same, but some, such as x_6, x_7, and x_8, have changed. I'm not entirely sure why this.

3 Question 3

Below, I simulated data that have a non-linear relationship.

```r
set.seed(7272)
n <- 250
dat <- tibble(x = rnorm(n, mean = 1, sd = 0.2)) |>
  mutate(y = exp(2.9 + 1.8*x + rnorm(n, mean = 0, sd = 1)))
```

3.1 Part a

Fit the model $y_i = \beta_0 + \beta_1 x_{1i} + \epsilon_i$ and interpret the coefficient on x.

```r
exp_mod <- lm(y ~ x, data = dat)
summary(exp_mod)
```

```
Call:
lm(formula = y ~ x, data = dat)

Residuals:
    Min      1Q  Median      3Q     Max
-209.57 -109.57  -46.30   39.57 1750.50

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   -26.45      66.85  -0.396  0.69269
x             190.28      65.99   2.883  0.00428 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

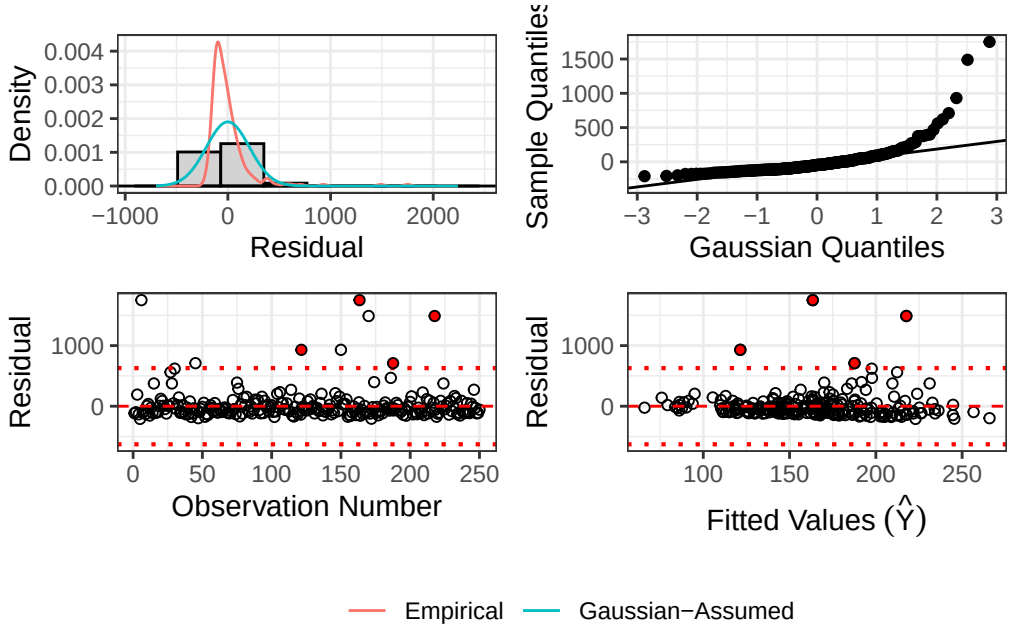
Residual standard error: 209.6 on 248 degrees of freedom
Multiple R-squared:  0.03243,   Adjusted R-squared:  0.02853
F-statistic: 8.313 on 1 and 248 DF,  p-value: 0.004281
```

From an OLS model, we expect that for every unit increase in x, y would increase by 190.28.

3.2 Part b

Conduct a residual analysis on the model in part a to demonstrate how to identify when you need a log transformation.

```
plotResiduals(exp_mod)
```



The residual analysis gives us several clues as to why a log transform would be helpful in this situation: mainly, the residuals are heavily right skewed with a huge tail, and all of the large residuals are positive. Also, the qq-plot shows a predictable deviance from the standard gaussian quantiles at high sample quantiles.

3.3 Part c

Fit the model $\log(y_i) = \beta_0 + \beta_1 x_{1i} + \epsilon_i$ and interpret the coefficient on x.

```
log_mod <- lm(log(y) ~ x, data = dat)
summary(log_mod)
```

```
Call:
lm(formula = log(y) ~ x, data = dat)

Residuals:
     Min       1Q   Median       3Q      Max
-2.35217 -0.61174  0.05022  0.67488  2.94090

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   3.5528     0.3046  11.664  < 2e-16 ***
x             1.0654     0.3007   3.543 0.000472 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

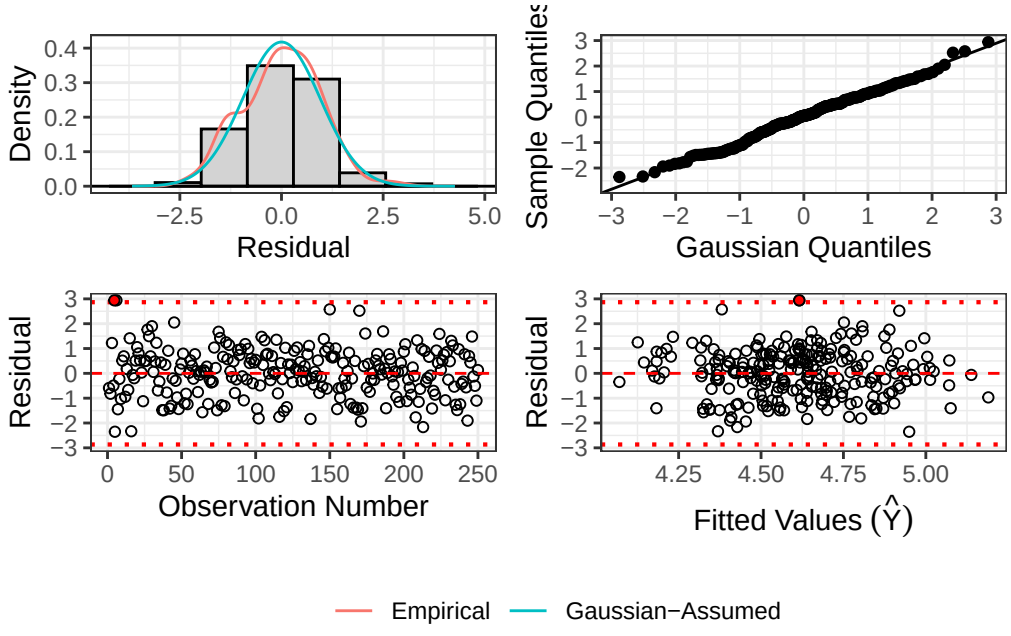
Residual standard error: 0.9552 on 248 degrees of freedom
Multiple R-squared:  0.04818,   Adjusted R-squared:  0.04434
F-statistic: 12.55 on 1 and 248 DF,  p-value: 0.0004723
```

Based on this log transformed model, for every unit increase in x, we'd expect a 1.0654 increase in $\log(y)$.

3.4 Part d

Conduct a residual analysis on the model in part c to demonstrate that the conditions for using regression are met.

```
plotResiduals(log_mod)
```



The log transformed model fits the Gaussian distribution much better now,the histogram is centered around 0, and it fits the qqplot almost perfectly. There's only residual that's slightly large enough to be noticeable among the rest of the residuals, but that's likely to happen with a sample size of 250.

3.5 Part e

Standardize x and fit the model $\log(y_i) = \beta_0 + \beta_1 z_{1i} + \epsilon_i$ and interpret the coefficient on z.

```
scaled_dat <- dat |>
  mutate(x = scale(x))

summary(lm(log(y) ~ x, data = dat))
```

```
Call:
lm(formula = log(y) ~ x, data = dat)

Residuals:
     Min       1Q   Median       3Q      Max
-2.35217 -0.61174  0.05022  0.67488  2.94090

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   3.5528     0.3046  11.664  < 2e-16 ***
x             1.0654     0.3007   3.543 0.000472 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.9552 on 248 degrees of freedom
Multiple R-squared:  0.04818,   Adjusted R-squared:  0.04434
F-statistic: 12.55 on 1 and 248 DF,  p-value: 0.0004723
```

The standardized and log transformed model shows us that for a standard deviation increase in x, we'd expect a 1.0654 increase in $\log(y)$.

4 AI Disclosure

I used Claude AI throughout to help write code and generally to remember how to use EMM packages and functions. The conversation of everything used can be found here.

References

Open Science Collaboration. 2015. "Estimating the Reproducibility of Psychological Science." *Science* 349 (6251): aac4716.